Integração Contínua, DevOps e Computação em Nuvem

Autor: Tiago Garcia Ferreira

Caso de Uso: Realizar o deploy de uma aplicação Node.js e Angular na AWS EKS, utilizando GitHub Actions como pipeline CI/CD, além dos serviços de PostgreSQL, Prometheus e Grafana.

Projeto: Disponível aqui

Aplicação desenvolvida para deploy

A aplicação consiste em um gerenciador de tarefas simples, desenvolvido com Node.js (backend) e Angular (frontend), com foco em operações básicas de produtividade. Essas operações seguem o padrão **CRUD**, com persistência de dados em um banco **PostgreSQL**, garantindo integridade e armazenamento estruturado das informações. A autenticação é realizada via **Firebase Authentication**, utilizando o **provider Google**, permitindo login seguro e simplificado sem necessidade de gerenciamento manual de credenciais.

A stack tecnológica completa, incluindo ferramentas de observabilidade, infraestrutura e CI/CD, pode ser consultada aqui.

Arquivos K8s - AWS EKS

Configurações gerais do Cluster

Configurações namespace app

Configurações namespace monitoring

Configurações deployment

Configurações postgres

Configurações prometheus

Configurações grafana

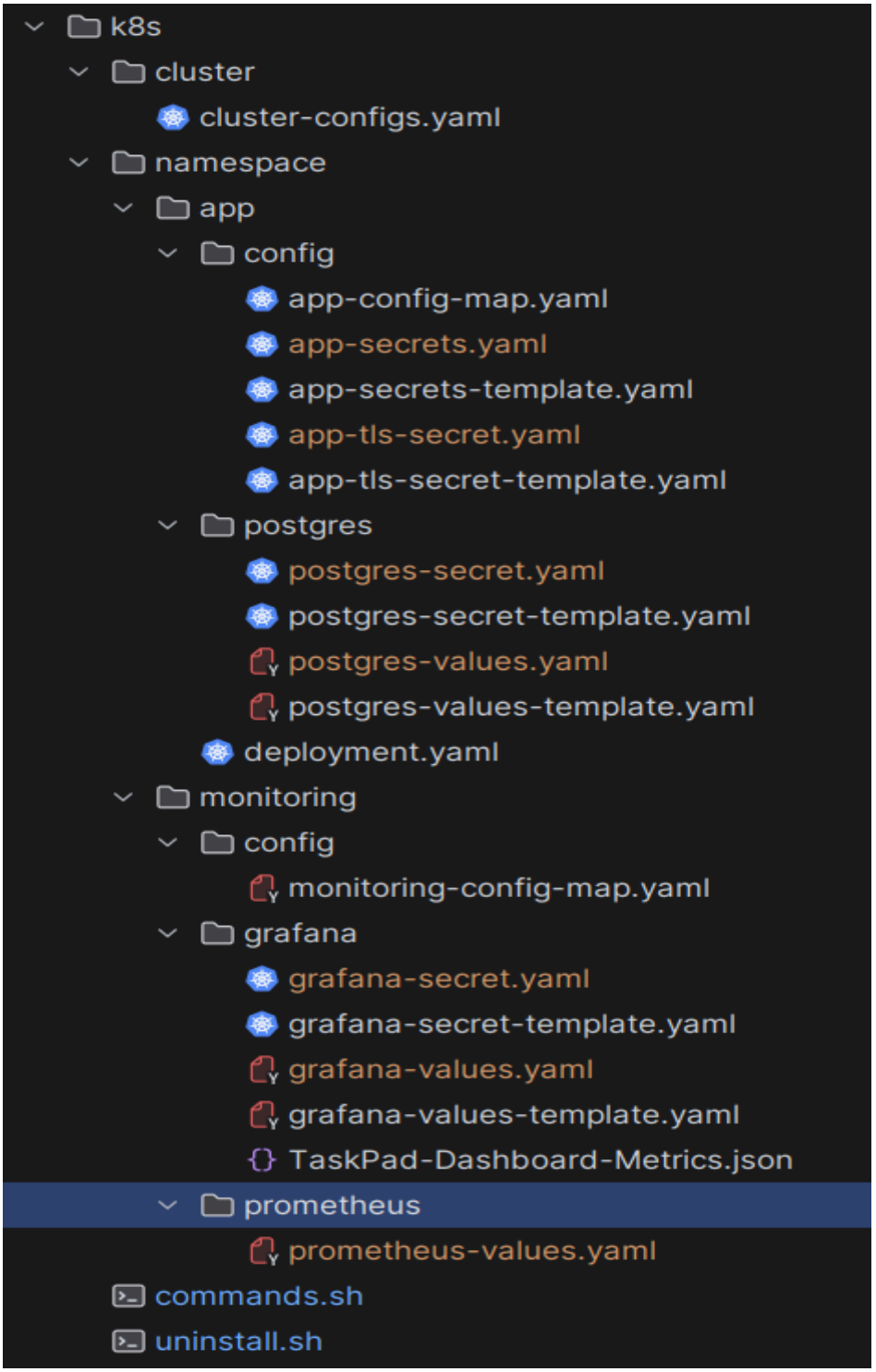
Configuração github actions

Comandos de instalação

Instalações:

Deploy de: PostgreSQL + Prometheus + Grafana realizado via Helm.

Deploy da Aplicação realizado via deployment.



Utilize o Docker para criar uma imagem personalizada de alguma aplicação previamente feita por você.

A. Publique a sua imagem no Docker Hub. [Docker Hub]

```
tgf@policorp:~/GitHub/task-pad$ make build
🔨 Building image...
sha256:de4953b052b17cdf5520c8b6ddb92fdd8c41e3fb0b907c5c6bfd88b8c0c06716
✅ Image built
```

```
tgf@policorp:~/GitHub/task-pad$ make tag
🏷️Tagging image...
docker tag tiagogferreirainfnet/task-pad:latest tiagogferreirainfnet/task-pad:1.3.0
✅ Tagged as: tiagogferreirainfnet/task-pad:1.3.0
```

```
tgf@policorp:~/GitHub/task-pad$ make push
📤 Pushing image to registry...
1.3.0: digest: sha256:de4953b052b17cdf5520c8b6ddb92fdd8c41e3fb0b907c5c6bfd88b8c0c06716 size: 856
latest: digest: sha256:de4953b052b17cdf5520c8b6ddb92fdd8c41e3fb0b907c5c6bfd88b8c0c06716 size: 856
✅ Image pushed
```

```
tgf@policorp:~/GitHub/task-pad$ make signature
🔐 Signing Docker image by digest...
📋 Digest: tiagogferreirainfnet/task-pad@sha256:de4953b052b17cdf5520c8b6ddb92fdd8c41e3fb0b907c5c6bfd88b8c0c06716
cosign sign --key .cosign/cosign.key tiagogferreirainfnet/task-pad@sha256:de4953b052b17cdf5520c8b6ddb92fdd8c41e3fb0b907c5c6bfd88b8c0c06716
✅ Image signed: tiagogferreirainfnet/task-pad:latest
```

```
tgf@policorp:~/GitHub/task-pad$ make info
📊 Image Information:
REPOSITORY:TAG                          SIZE      CREATED AT
tiagogferreirainfnet/task-pad:1.3.0     219MB     2026-04-13 12:11:03 -0300 -03
tiagogferreirainfnet/task-pad:latest    219MB     2026-04-13 12:11:03 -0300 -03
  📦        Name: tiagogferreirainfnet/task-pad
  🏷️        Tag: latest - 1.3.0
  🔐   Signature: ✅ Signed (Docker Trust)
```

Repositories / task-pad / Tags / latest

tiagogferreirainfnet/task-pad:latest

API MANAGEMENT | DEVELOPER TOOLS | WEB SERVERS

INDEX DIGEST sha256:de4953b052b17cdf5520c8b6ddb92fdd8c41e3fb0b907c5c6bfd88b8c0c06716

OS/ARCH	COMPRESSED SIZE	LAST PUSHED	TYPE	MANIFEST DIGEST
linux/amd64	53.46 MB	4 minutes by tiagogferreirainfnet	Image	sha256:75da9cb8...

[Comandos] - [Dockerfile]

Suba sua imagem em algum cluster kubernetes, seguindo as seguintes especificações.

A. Utilize Deployment para subir sua aplicação com 4 réplicas.

```yaml
deployment.yaml
apiVersion: apps/v1
kind: Deployment
metadata:
  name: taskpad-app
  namespace: app
  labels:
    app: taskpad-app
spec:
  replicas: 4
  selector:
    matchLabels:
      app: taskpad-app
```

```
tgf@policorp:~/GitHub/task-pad/k8s$ kubectl get pods --namespace app
NAME                                READY   STATUS    RESTARTS   AGE
taskpad-app-8d88c75db-fsn9l         1/1     Running   0          18m
taskpad-app-8d88c75db-td66z         1/1     Running   0          11s
taskpad-app-8d88c75db-wgf6t         1/1     Running   0          11s
taskpad-app-8d88c75db-zdqwm         1/1     Running   0          11s
taskpad-postgres-postgresql-0       1/1     Running   0          66m
```

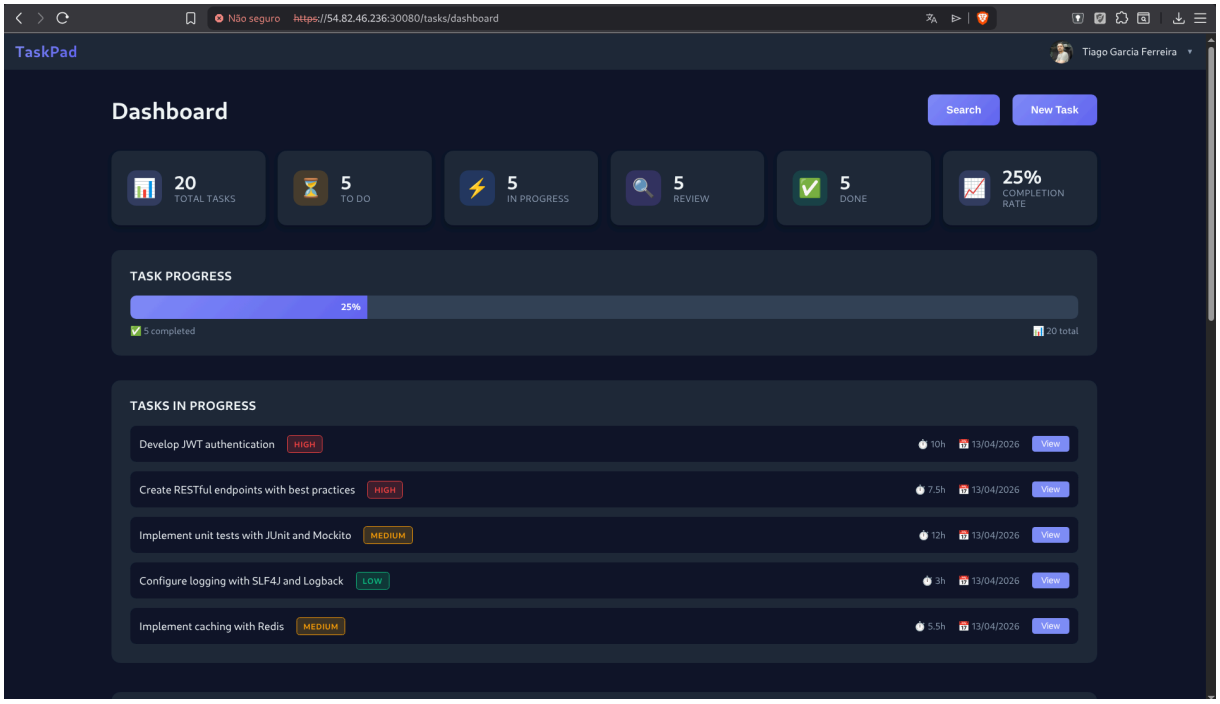
[Deployment]

B. Exponha sua aplicação de modo que ela fique acessível fora do Cluster.

```yaml
apiVersion: v1
kind: Service
metadata:
  name: taskpad-app-service
  namespace: app
  labels:
    app: taskpad-app
  annotations: 4 items
spec:
  type: NodePort
  selector:
    app: taskpad-app
  ports:
    - name: https
      port: 4000
      targetPort: 4000
      nodePort: 30080
```

```
tgf@policorp:~/GitHub/task-pad/k8s$ kubectl get svc taskpad-app-service -n app
NAME                 TYPE       CLUSTER-IP      EXTERNAL-IP   PORT(S)          AGE
taskpad-app-service  NodePort   10.100.197.40   <none>        4000:30080/TCP   35m
```

```
tgf@policorp:~/GitHub/task-pad/k8s$ kubectl get nodes -o jsonpath='{range .items[*]}{.status.addresses[?(@.type=="ExternalIP")].address}{"\n"}{end}'
100.48.17.9
3.239.122.40
54.82.46.236
```



Deployment -> **kind: *Service***

C. Se sua aplicação fizer uso de banco de dados, crie um POD com o mesmo e deixe-o acessível através do ClusterIP. Se sua aplicação não fizer uso de um BD suba uma imagem do Redis e crie um ClusterIP para o mesmo.

```yaml
architecture: "standalone"

auth:
  existingSecret: "postgres-secret"
  database: "taskpad"
  username: "taskpad_prod"
  secretKeys:
    userPasswordKey: "password"
    postgresPasswordKey: "postgres-password"

primary: {2 keys}

service:
  type: ClusterIP
  port: "5432"

metrics: {1 key}
```

```
tgf@policorp:~/GitHub/task-pad/k8s$ kubectl get pvc --namespace app
NAME                                STATUS   VOLUME                                     CAPACITY   ACCESS MODES   STORAGECLASS   VOLUMEATTRIBUTESCLASS   AGE
data-taskpad-postgres-postgresql-0  Bound    pvc-d5745397-69de-4830-a9a1-e9d68e3b96d4   2Gi        RWO            gp3            <unset>                 61m
```

```
tgf@policorp:~/GitHub/task-pad/k8s$ kubectl get pods --namespace app
NAME                                READY   STATUS    RESTARTS   AGE
taskpad-app-8d88c75db-fsn9l         1/1     Running   0          26m
taskpad-postgres-postgresql-0       1/1     Running   0          75m
```

```
tgf@policorp:~/GitHub/task-pad/k8s$ kubectl get svc taskpad-postgres-postgresql -n app
NAME                          TYPE        CLUSTER-IP      EXTERNAL-IP   PORT(S)    AGE
taskpad-postgres-postgresql   ClusterIP   10.100.143.12   <none>        5432/TCP   76m
```

```
tgf@policorp:~/GitHub/task-pad/k8s$ kubectl get svc taskpad-app-service -n app
NAME                 TYPE       CLUSTER-IP      EXTERNAL-IP   PORT(S)          AGE
taskpad-app-service  NodePort   10.100.197.40   <none>        4000:30080/TCP   35m
```

Postgres

D. Crie algum probe para sua aplicação (Readness ou Liveness.)

```yaml
startupProbe:
  httpGet:
    path: /live
    port: 4000
    scheme: HTTPS
  periodSeconds: 10
  timeoutSeconds: 3
  successThreshold: 1
  failureThreshold: 3
livenessProbe:
  httpGet:
    path: /live
    port: 4000
    scheme: HTTPS
  periodSeconds: 10
  timeoutSeconds: 5
  successThreshold: 1
  failureThreshold: 3
readinessProbe:
  httpGet:
    path: /ready
    port: 4000
    scheme: HTTPS
  periodSeconds: 10
  timeoutSeconds: 5
  successThreshold: 1
  failureThreshold: 3
```

Crie a estrutura para monitorar sua aplicação com o Prometheus e o Grafana (ou qualquer ferramenta a sua escolha: você deve ter um servidor de métricas e alguma ferramenta para dashboards).

a. Apenas o Grafana deverá ficar acessível para fora do Cluster.

Prometheus

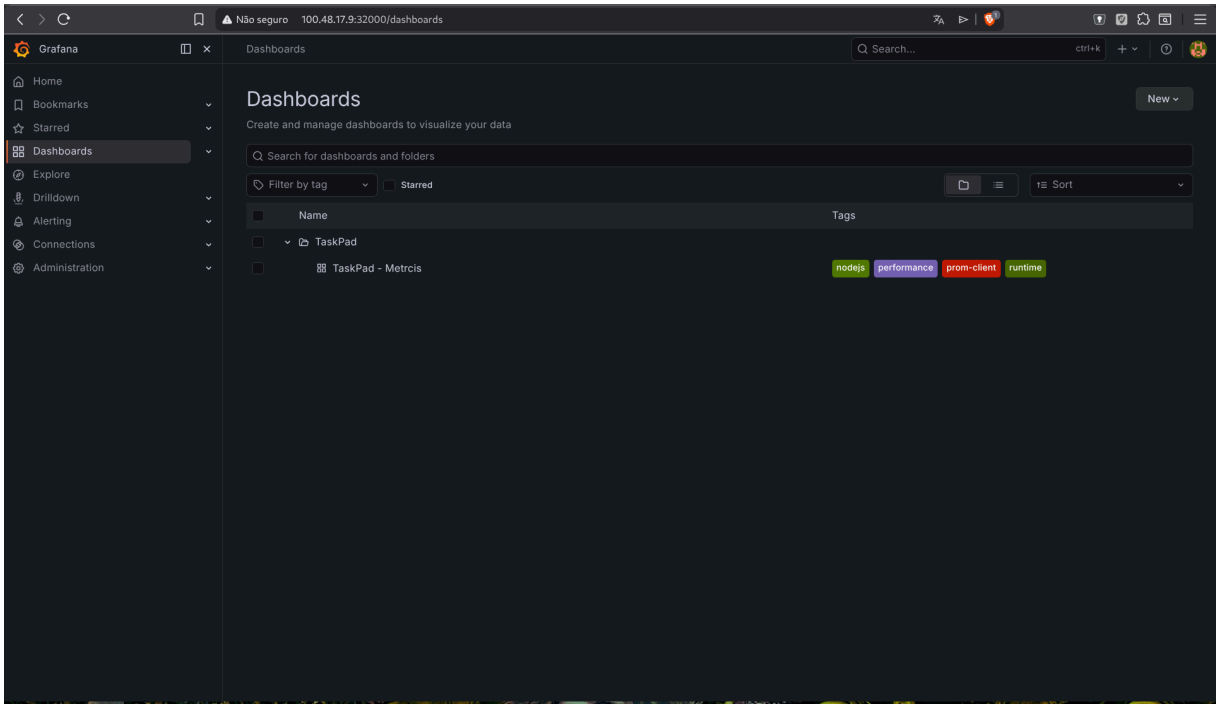
```
service:
  type: ClusterIP
```

```
tgf@policorp:~/GitHub/task-pad/k8s$ kubectl get svc  -n monitoring
NAME                        TYPE        CLUSTER-IP      EXTERNAL-IP   PORT(S)   AGE
taskpad-prometheus-server   ClusterIP   10.100.10.201   <none>        80/TCP    2m6s
```

Grafana

```
service:
  type: NodePort
  nodePort: 32000
```

```
tgf@policorp:~/GitHub/task-pad/k8s$ kubectl get svc --namespace monitoring
NAME                        TYPE        CLUSTER-IP       EXTERNAL-IP   PORT(S)        AGE
taskpad-grafana             NodePort    10.100.235.199   <none>        80:32000/TCP   9m4s
```

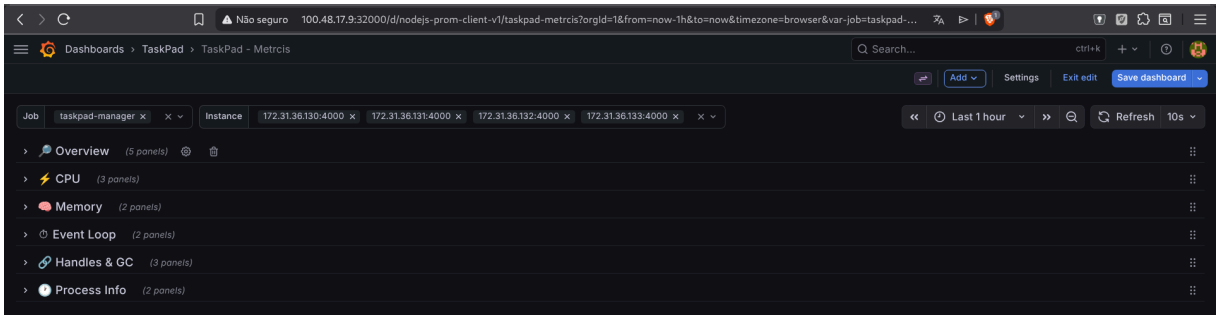


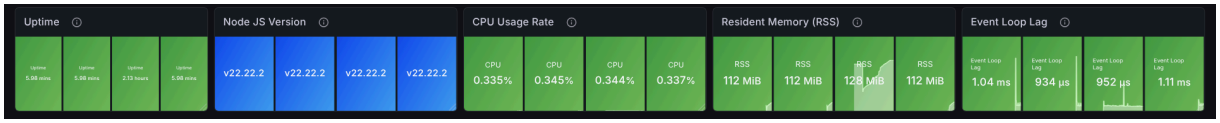
b. Utilize um PVC para escrever os dados do Prometheus de maneira persistente.

```
25      retention: "24h"
26      retentionSize: "2GB"
27
28      persistentVolume:
29        enabled: true
30        storageClass: "gp3"
31        size: "2Gi"
```

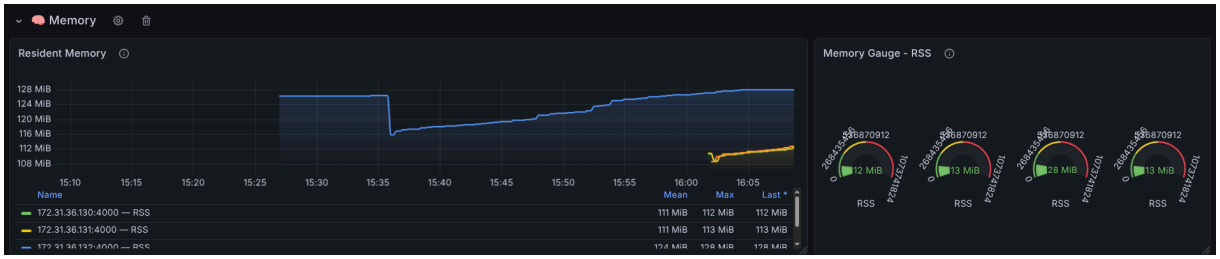
```
tgf@policorp:~/GitHub/task-pad/k8s$ kubectl get pvc --namespace monitoring
NAME                      STATUS   VOLUME                                     CAPACITY   ACCESS MODES   STORAGECLASS   VOLUMEATTRIBUTESCLASS   AGE
taskpad-prometheus-server Bound    pvc-064dfcb1-311f-4947-968b-8bef26e86039   2Gi        RWO            gp3            <unset>                 5m47s
```

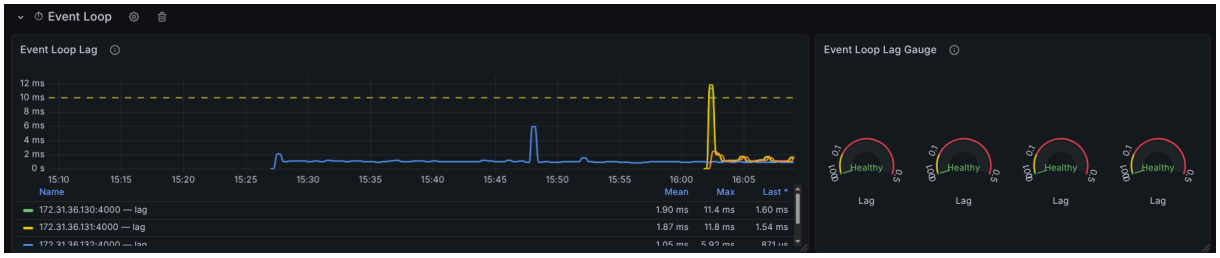
c. Crie dashboards do Grafana que exponha dados sensíveis da sua aplicação (memória, cpu, etc.)

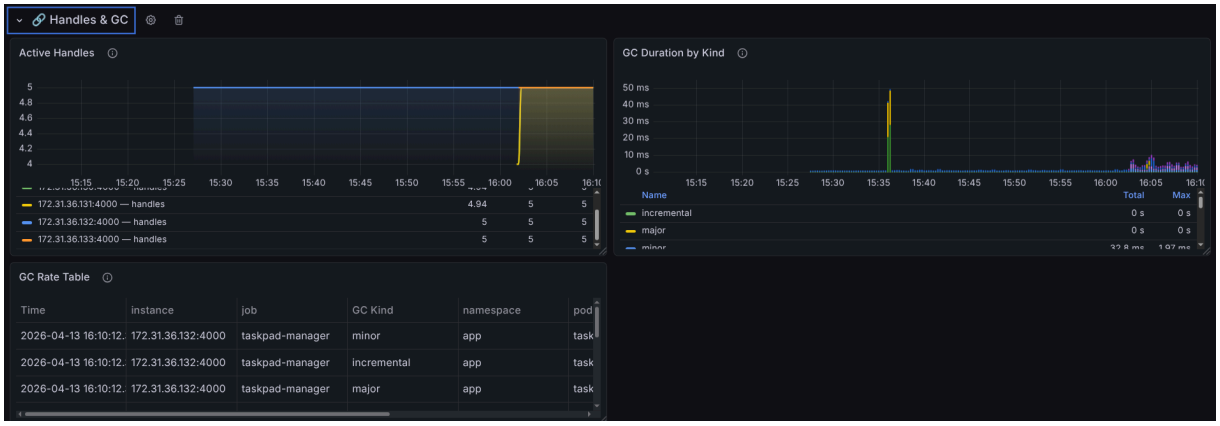


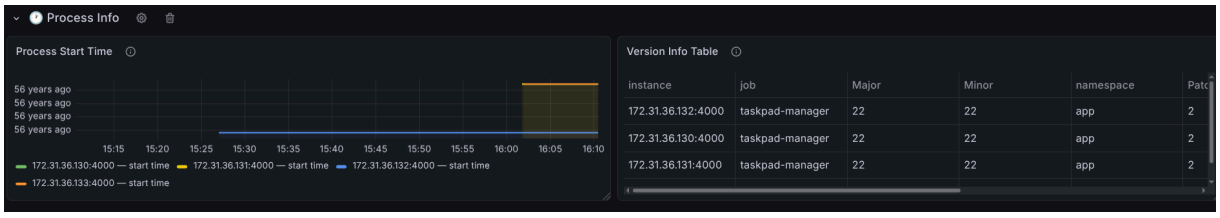

Uptime
Node JS Version
v22.22.2
v22.22.2
v22.22.2
v22.22.2
CPU Usage Rate
CPU 0.335%
CPU 0.345%
CPU 0.344%
CPU 0.337%
Resident Memory (RSS)
RSS 112 MiB
RSS 112 MiB
RSS 128 MiB
RSS 112 MiB
Event Loop Lag
Event Loop Lag 1.04 ms
Event Loop Lag 934 µs
Event Loop Lag 952 µs
Event Loop Lag 1.11 ms

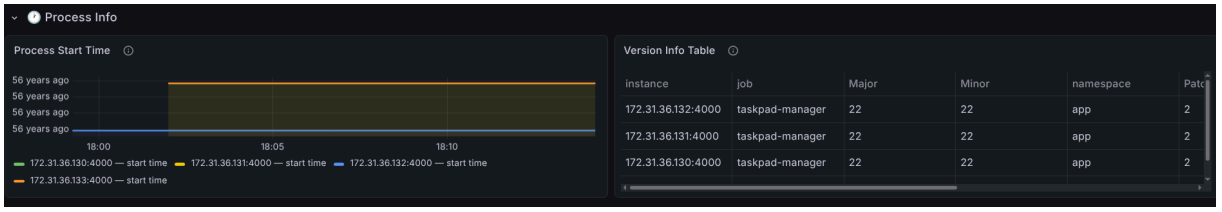
CPU
CPU Usage Rate
0.7%
0.6%
0.5%
0.4%
0.3%
15:10 15:15 15:20 15:25 15:30 15:35 15:40 15:45 15:50 15:55 16:00 16:05
Name Mean Max Last *
172.31.36.130:4000 — CPU rate 0.372% 0.602% 0.353%
172.31.36.131:4000 — CPU rate 0.377% 0.591% 0.371%
CPU Gauge
0.353% CPU
0.371% CPU
0.359% CPU
0.374% CPU
CPU Accumulated
CPU seconds total 1.56 s
CPU seconds total 1.59 s
CPU seconds total 31.2 s
CPU seconds total 1.66 s


Memory
Resident Memory
128 MiB
124 MiB
120 MiB
116 MiB
112 MiB
108 MiB
15:10 15:15 15:20 15:25 15:30 15:35 15:40 15:45 15:50 15:55 16:00 16:05
Name Mean Max Last *
172.31.36.130:4000 — RSS 111 MiB 112 MiB 112 MiB
172.31.36.131:4000 — RSS 111 MiB 113 MiB 113 MiB
Memory Gauge - RSS
12 MiB RSS
13 MiB RSS
28 MiB RSS
13 MiB RSS


Event Loop
Event Loop Lag
12 ms
10 ms
8 ms
6 ms
4 ms
2 ms
0 s
15:10 15:15 15:20 15:25 15:30 15:35 15:40 15:45 15:50 15:55 16:00 16:05
Name Mean Max Last *
172.31.36.130:4000 — lag 1.90 ms 11.4 ms 1.60 ms
172.31.36.131:4000 — lag 1.87 ms 11.8 ms 1.54 ms
Event Loop Lag Gauge
Healthy Lag
Healthy Lag
Healthy Lag
Healthy Lag

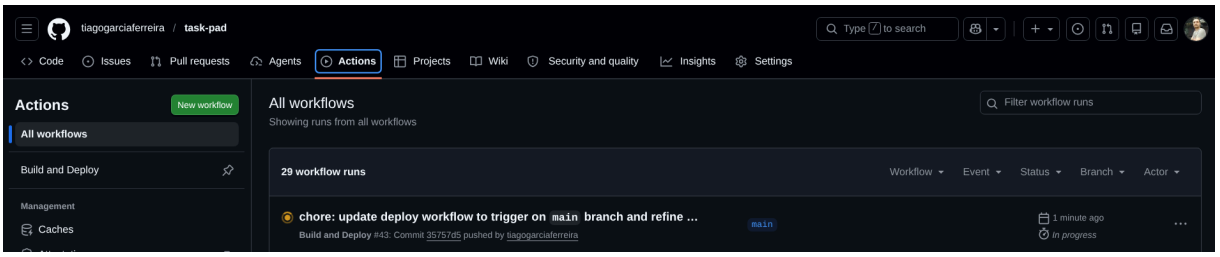

Handles & GC
Active Handles
5
4.8
4.6
4.4
4.2
4
172.31.36.131:4000 — handles 4.94 5 5
172.31.36.132:4000 — handles 5 5 5
172.31.36.133:4000 — handles 5 5 5
GC Duration by Kind
50 ms
40 ms
30 ms
20 ms
10 ms
0 s
15:15 15:20 15:25 15:30 15:35 15:40 15:45 15:50 15:55 16:00 16:05
Name Total Max
incremental 0 s 0 s
major 0 s 0 s
GC Rate Table
Time instance job GC Kind namespace pod
2026-04-13 16:10:12. 172.31.36.132:4000 taskpad-manager minor app task
2026-04-13 16:10:12. 172.31.36.132:4000 taskpad-manager incremental app task
2026-04-13 16:10:12. 172.31.36.132:4000 taskpad-manager major app task


Process Info
Process Start Time
56 years ago
56 years ago
56 years ago
56 years ago
15:15 15:20 15:25 15:30 15:35 15:40 15:45 15:50 15:55 16:00 16:05
172.31.36.130:4000 — start time
172.31.36.131:4000 — start time
172.31.36.132:4000 — start time
172.31.36.133:4000 — start time
Version Info Table
instance job Major Minor namespace
172.31.36.132:4000 taskpad-manager 22 22 app 2
172.31.36.130:4000 taskpad-manager 22 22 app 2
172.31.36.131:4000 taskpad-manager 22 22 app 2



Utilize o Jenkins (ou qualquer ferramenta) para criar um pipeline de entrega do seu projeto.

```yaml
on:
  push:
    branches: ['main']
    paths-ignore:
      - 'k8s/**'
      - '.angular/**'
      - '.editorconfig'
      - '.prettierrc'
      - '.gitignore'
      - '.env.template'
      - '.github/**'
      - 'dist/**'
      - 'docker/**'
      - 'tasks.json'
      - 'proxy.conf.js'
      - 'Makefile'
      - 'README.md'
      - 'k6/**'
```



tiagogarciaferreira / task-pad

Code | Issues | Pull requests | Agents | Actions

Para sair do modo tela cheia, toque em Esc e pressione

Build and Deploy

chore: update deploy workflow to trigger on `main` branch and refine ... #43

Cancel workflow

Summary

All jobs

build

Run details

Usage

Workflow file

Triggered via push 1 minute ago

tiagogarciaferreira pushed 35757d5 main

Status: In progress

Total duration: –

Artifacts: –

deploy.yml

on: push

build 1m 49s → deploy

---

tiagogarciaferreira / task-pad

Code | Issues | Pull requests | Agents | Actions

Para sair do modo tela cheia, toque em Esc e pressione

Build and Deploy

chore: update deploy workflow to trigger on `main` branch and refine ... #43

Cancel workflow

Summary

All jobs

build

deploy

Run details

Usage

Workflow file

Triggered via push 5 minutes ago

tiagogarciaferreira pushed 35757d5 main

Status: In progress

Total duration: –

Artifacts: 1

deploy.yml

on: push

build 4m 53s → deploy 8s

Build and Deploy

chore: update deploy workflow to trigger on `main` branch and refine ... #43

Re-run all jobs | Latest #2

Summary

All jobs

build

deploy

Run details

Usage

Workflow file

build

succeeded 9 minutes ago in 4m 53s

Search logs

Set up job	5s
Checkout code	1s
Setup Node.js	8s
Cache node_modules	0s
Install dependencies	22s
Install Angular CLI	18s
Create environment files	0s
Cache Angular build cache	0s
Build application	11s
Set up Docker Buildx	7s
Log in to Docker Hub	0s
Get version	0s
Extract metadata for Docker	1s
Build and push Docker image	2m 59s
Dive analyze	7s
Install Cosign	3s
Decode Cosign signing key	0s
Sign Docker image	9s
Cleanup signing key	0s

Build and push Docker image 2m 59s

```
1   ▶ Run docker/build-push-action@v7
50  ▶ GitHub Actions runtime token ACs
52  ▶ Docker info
143 ▶ Proxy configuration
145 ▶ Buildx version
148 ▶ Builder info
211 /usr/bin/docker buildx build --build-arg NODE_ENV=production --build-arg BUILD_MODE=production --cache-from type=gha --cache-to type=gha,mode=max --file
    docker/Dockerfile --iidfile /home/runner/work/_temp/docker-actions-toolkit-iDayja/build-iidfile-a4414dc14d.txt --attest type=provenance,mode=min,inline-
    only=true,builder-id=https://github.com/tiagogarciaferreira/task-pad/actions/runs/24368585402/attempts/1 --tag ***/task-pad:latest --tag ***/task-pad:1.3.0 --tag
    ***/task-pad:sha-35757d53a8b4b7d65be81c44b837638fa12ecf45 --metadata-file /home/runner/work/_temp/docker-actions-toolkit-iDayja/build-metadata-8b500b3396.json --
    push .
212 #0 building with "builder-73c9f706-18a4-4e9a-ae0d-091ea5c1f0f2" instance using docker-container driver
```

Repositories / task-pad / Tags

tiagogferreirainfnet/task-pad

Last pushed 13 minutes ago · Repository size: 7.2 GB · 0 · 7.4K

TaskPad - Task Management System

API MANAGEMENT | DEVELOPER TOOLS | WEB SERVERS

General | Tags | Image Management | Collaborators | Webhooks | Settings

Docker commands

Public view

To push a new tag to this repository:

```
docker push tiagogferreirainfnet/task-pad:tagname
```

Sort by Newest | sha-35757d53a8b4b7d6! | Delete

TAG

sha-35757d53a8b4b7d65be81c44b837638fa12ecf45

Last pushed 16 minutes by tiagogferreirainfnet

```
docker pull tiagogferreirainfnet/task-pad:sha-35757d53a8b4b7d65be81c44b837638fa12ecf45
```

Digest	OS/ARCH	Last pull	Compressed size
71f461aa8836	linux/amd64	less than 1 day	53.46 MB

tiagogarciaferreira / task-pad

Type / to search

Code | Issues | Pull requests | Agents | Actions | Projects | Wiki | Security and quality | Insights | Settings

Build and Deploy

chore: update deploy workflow to trigger on `main` branch and refine ... #43

Re-run all jobs | Latest #2

Summary

All jobs

- build
- deploy

Run details

- Usage
- Workflow file

Annotations
3 warnings

deploy
succeeded 4 minutes ago in 47s

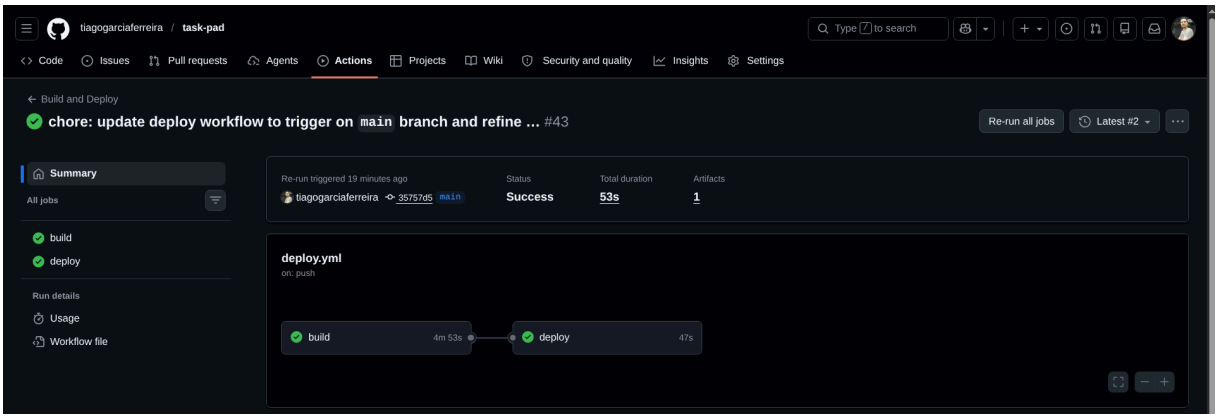
Search logs

Set up job	2s
Generate token do App	1s
Checkout code	1s
Configure AWS credentials	1s
Install kubectl	0s
Update kubeconfig	4s
Extract the current image	1s
Install Cosign	3s
Decode Cosign signing key	0s
Verify Docker image signature	5s
Deploy on EKS	23s
Update image in deployment	1s

Deploy on EKS 23s

```
1  ▶ Run kubectl set image deployment/taskpad-app taskpad-app="$DOCKER_HUB_REPO@$NEW_IMAGE_DIGEST" -n app
14 deployment.apps/taskpad-app image updated
15 Waiting for deployment "taskpad-app" rollout to finish: 2 out of 4 new replicas have been updated...
16 Waiting for deployment "taskpad-app" rollout to finish: 2 out of 4 new replicas have been updated...
17 Waiting for deployment "taskpad-app" rollout to finish: 2 out of 4 new replicas have been updated...
18 Waiting for deployment "taskpad-app" rollout to finish: 2 out of 4 new replicas have been updated...
19 Waiting for deployment "taskpad-app" rollout to finish: 3 out of 4 new replicas have been updated...
20 Waiting for deployment "taskpad-app" rollout to finish: 3 out of 4 new replicas have been updated...
21 Waiting for deployment "taskpad-app" rollout to finish: 1 old replicas are pending termination...
22 Waiting for deployment "taskpad-app" rollout to finish: 1 old replicas are pending termination...
23 Waiting for deployment "taskpad-app" rollout to finish: 1 old replicas are pending termination...
24 deployment "taskpad-app" successfully rolled out
```

```
Update image in deployment                                                    1s
▼ Run FILE="k8s/namespace/app/deployment.yaml"
  FILE="k8s/namespace/app/deployment.yaml"
  sed -i "s|$OLD_IMAGE_DIGEST|$NEW_IMAGE_DIGEST|g" "$FILE"
  git config user.name "app-auto-commit-bot[bot]"
  git config user.email "app-auto-commit-bot@github.com"
  git add "$FILE"
  git commit -m "chore: update image to $NEW_IMAGE_DIGEST"
  git push
  shell: /usr/bin/bash -e {0}
  env:
    DOCKER_HUB_REPO: ***/task-pad
    NEW_IMAGE_DIGEST: sha256:73f78637d179153b22a2b63852b7a2b70a52b80a5b7292997d9e4e23a71b01a3
    AWS_DEFAULT_REGION: us-east-1
    AWS_REGION: us-east-1
    AWS_ACCESS_KEY_ID: ***
    AWS_SECRET_ACCESS_KEY: ***
    AWS_SESSION_TOKEN: ***
    OLD_IMAGE_DIGEST: sha256:de4953b052b17cdf5520c8b6ddb92fdd8c41e3fb0b907c5c6bfd88b8c0c06716
[main 5942a79] chore: update image to sha256:73f78637d179153b22a2b63852b7a2b70a52b80a5b7292997d9e4e23a71b01a3
 1 file changed, 1 insertion(+), 1 deletion(-)
remote: Bypassed rule violations for refs/heads/main:
remote:
remote: - Changes must be made through a pull request.
remote:
remote: - Commits must have verified signatures.
remote:   Found 1 violation:
remote:
remote:   5942a795fcacad0fcd8ef1f7934b003de40133b8
remote:
To https://github.com/tiagogarciaferreira/task-pad
   35757d5..5942a79  main -> main
```



Antes do Workflow

```
tgf@policorp:~/GitHub/task-pad$ kubectl get deployment taskpad-app -n app -o jsonpath='{.spec.template.spec.containers[0].image}{"\n"}'
tiagogferreirainfnet/task-pad@sha256:de4953b052b17cdf5520c8b6ddb92fdd8c41e3fb0b907c5c6bfd88b8c0c06716
tgf@policorp:~/GitHub/task-pad$ ~
```

```
21        containers:
22          - name: taskpad-app
23            image: tiagogferreirainfnet/task-pad@sha256:de4953b052b17cdf5520c8b6ddb92fdd8c41e3fb0b907c5c6bfd88b8c0c06716
24            imagePullPolicy: IfNotPresent
```

Depois do Workflow

```
tgf@policorp:~/GitHub/task-pad$ kubectl get deployment taskpad-app -n app -o jsonpath='{.spec.template.spec.containers[0].image}{"\n"}'
tiagogferreirainfnet/task-pad@sha256:73f78637d179153b22a2b63852b7a2b70a52b80a5b7292997d9e4e23a71b01a3
tgf@policorp:~/GitHub/task-pad$
```

```
        containers:
          - name: taskpad-app
            image: tiagogferreirainfnet/task-pad@sha256:73f78637d179153b22a2b63852b7a2b70a52b80a5b7292997d9e4e23a71b01a3
            imagePullPolicy: IfNotPresent
```

Execute um stress teste do seu projeto e tire print do Dashboard sofrendo alterações.

O teste foi realizado na página de login com 100 usuários por 2 minutos e depois por 5 minutos. [Script K6]

```
tgf@policorp:~/GitHub/task-pad$ k6 run --insecure-skip-tls-verify k6/tests/login-page.js

       /\      Grafana   /‾‾/
    /\  /  \     |\  __   /  /
   /  \/    \    | |/ /  /   ‾‾\
  /          \   |  (  |  (‾)  |
 / __________ \  |_|\_\  \_____/


     execution: local
        script: k6/tests/login-page.js
        output: -

     scenarios: (100.00%) 1 scenario, 100 max VUs, 5m30s max duration (incl. graceful stop):
              * default: 100 looping VUs for 5m0s (gracefulStop: 30s)


  TOTAL RESULTS

    checks_total.......: 77241   256.476964/s
    checks_succeeded...: 100.00% 77241 out of 77241
    checks_failed......: 0.00%   0 out of 77241

    ✓ status is 200

    HTTP
    http_req_duration..............: avg=161.62ms min=152.56ms med=160.09ms max=1.66s p(90)=163.75ms p(95)=166.11ms
      { expected_response:true }...: avg=161.62ms min=152.56ms med=160.09ms max=1.66s p(90)=163.75ms p(95)=166.11ms
    http_req_failed................: 0.00%  0 out of 77241
    http_reqs......................: 77241  256.476964/s

    EXECUTION
    iteration_duration.............: avg=1.16s    min=1.15s    med=1.16s    max=2.66s p(90)=1.16s    p(95)=1.16s
    iterations.....................: 25747  85.492321/s
    vus............................: 4      min=4          max=100
    vus_max........................: 100    min=100        max=100

    NETWORK
    data_received..................: 181 MB 600 kB/s
    data_sent......................: 8.2 MB 27 kB/s
```

Event Loop
Event Loop Lag
5 ms
4 ms
3 ms
2 ms
1 ms
18:10
18:15
18:20
18:25
18:30
18:35
Name
Mean
Max
Last *
172.31.36.130:4000 — lag
991 µs
2.17 ms
810 µs
172.31.36.131:4000 — lag
1.00 ms
4.99 ms
761 µs

Handles & GC
Active Handles
80
60
40
20
0
18:10
18:15
18:20
18:25
18:30
18:35
Name
Mean
Max
Last *
172.31.36.130:4000 — handles
24.0
79
5
172.31.36.131:4000 — handles
26.3
82
5
GC Duration by Kind
1.4 s
1.2 s
1 s
800 ms
600 ms
400 ms
200 ms
0 s
18:10
18:15
18:20
18:25
18:30
18:35
Name
Total
Max
incremental
307 ms
23.6 ms
major
991 ms
52.7 ms